

Political Bias and Temporal Dynamics in Large Language Models

State of the Art

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Outline

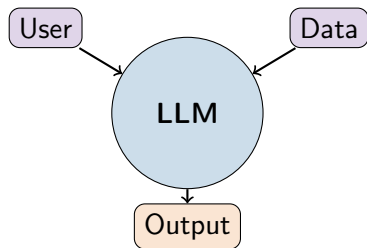
- 1 Introduction
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Large Language Models are now **central intermediaries** for:

- Accessing political information
- Interpreting complex topics
- Evaluating political claims
- Content summarization

! Key Question:

Are these models ideologically neutral?



Two Research Directions



Political Bias in LLMs

How do LLMs express systematic evaluative patterns when discussing political topics?

Key concerns:

- Ideological preferences
- Bias amplification
- Digital soft power



Temporal Dynamics

How do LLMs handle time-sensitive information and evolve over time?

Key concerns:

- Temporal generalization
- Performance decay
- Knowledge staleness

This survey examines the intersection of these two areas.

What is Political Bias in LLMs?

Definition (Buyl et al., 2026)

Ideology: A systematic worldview concerning social organization, power distribution, and moral evaluation.

How political bias manifests in LLMs:

- **Evaluative discourse** — How models morally frame political actors
- **Sentiment patterns** — Positive/negative treatment of political labels
- **Moral framing** — How movements and doctrines are presented

Key Finding

LLM behavior is shaped by: (1) architectural choices, (2) training data composition, and (3) post-training alignment procedures.

Bias Amplification (Zhu et al., 2024)

Biases present in training data are not only *reproduced* but potentially *intensified* in model outputs, reflecting and exacerbating existing societal prejudices.

! This makes LLMs potential **instruments of digital soft power** when deployed at scale.

Two-Step Prompting Strategy

- ➊ **Step 1:** Elicit descriptive text about political topic
- ➋ **Step 2:** Ask model to self-assess moral stance (Likert scale)

Metrics used:

- Sentiment scores
- Moral evaluation scores
- Preferences toward political labels

✓ Strengths:

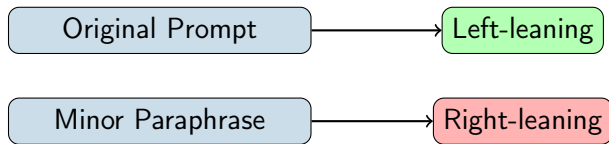
- Large-scale cross-model comparison
- Systematic evaluation

✗ Limitations:

- Relies on model introspection
- May miss implicit biases

Critical Finding

Multiple-choice political surveys (e.g., Political Compass Test) are **highly unstable** for evaluating LLMs!



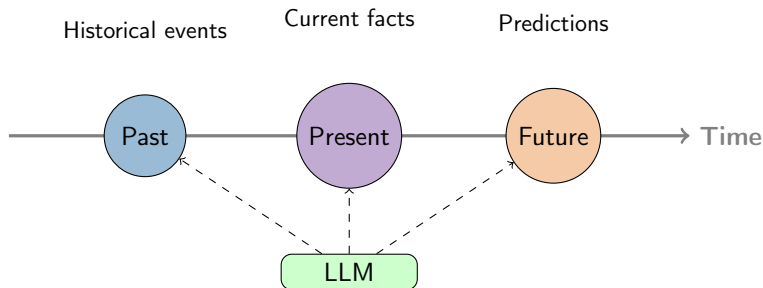
Same model, dramatically different results!

Implication: Raises concerns about **validity and robustness** of forced-choice political evaluations.

Temporal Generalization

Definition (Zhu et al., 2024)

Temporal Generalization: A model's ability to reason accurately across past, present, and future contexts.



Challenge: LLMs must generalize across all temporal contexts accurately.

Key Temporal Bias Concepts

Nostalgia Bias

Over-reliance on **historical data** when generating text.

Example: "The current president is..." returns outdated information.

Neophilia Bias

Overemphasis on **novelty** and future trends.

Example: Overpredicting adoption of new technologies.

Temporal Degeneration

Decline in LLM's performance after release.

As time passes, the model's knowledge becomes increasingly **stale** and its accuracy degrades.

This is particularly problematic for time-sensitive domains like:

- News and current events
- Political information
- Scientific developments

FreshBench Framework

A benchmark specifically designed to evaluate **temporal robustness** of LLMs.

What it measures:

- Performance decay over time
- Ability to handle time-sensitive queries
- Temporal reasoning capabilities

✓ Strengths:

- Systematic temporal analysis
- Valuable for longitudinal studies

✗ Limitations:

- Focuses on robustness
- Not directly on ideology

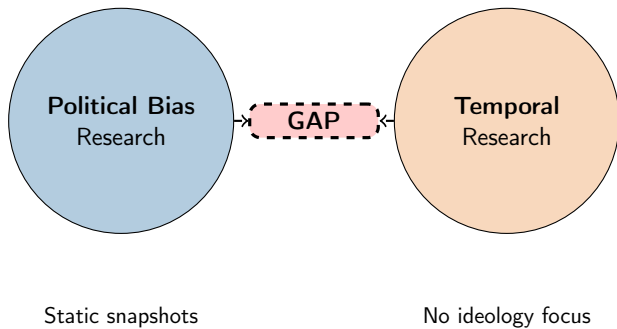
Summary of Methodologies

Study	Approach	Metrics	Focus
Buyl et al. (2026)	Two-step prompting + Likert self-assessment	Sentiment & moral scores	Political ideology
Röttger et al. (2024)	Critical examination of political tools	Survey stability	Evaluation validity
Zhu et al. (2024)	FreshBench temporal framework	Performance decay	Temporal robustness

Key Observation

These approaches are **complementary** but currently studied in isolation.

The Missing Link



Research Gap

Political bias is predominantly studied as a **static property**. But if LLMs degrade over time, political biases may **shift, intensify, or invert** across model versions.

Open Research Questions

- ➊ **Stability:** Are political biases *stable design features* or *evolving artifacts*?
- ➋ **Evolution:** How do biases change across model versions?
(e.g., GPT-3 → GPT-4 → GPT-5)
- ➌ **Temporal Effects:** Does temporal degeneration affect political stance?
- ➍ **Alignment Impact:** How do different alignment procedures affect bias trajectories?
- ➎ **Cross-lingual:** Do temporal-political bias patterns differ across languages?

What the Field Needs

Longitudinal studies that integrate both perspectives:

Methodological needs:

- Track bias across model versions
- Measure bias before/after events
- Compare alignment procedures
- Cross-model temporal analysis

Potential insights:

- Bias trajectories over time
- Effects of interventions
- Stability patterns
- Predictive indicators

Goal: Understand how political biases *evolve* in LLMs over time.

Key Takeaways

- 1 LLMs exhibit **systematic political biases** shaped by training and alignment
- 2 **Bias amplification** means training biases can be intensified in outputs
- 3 **Evaluation tools** (like Political Compass) show instability issues
- 4 **Temporal dynamics** (nostalgia, neophilia, degeneration) affect LLM behavior
- 5 **Critical gap:** Political bias and temporal dynamics are studied separately
- 6 **Future:** Longitudinal studies integrating both perspectives are needed

- Buyl, M., et al. (2026). *Large Language Models and Political Ideology*.
- Röttger, P., et al. (2024). *Political Compass Tests for LLMs: Critical Examination of Evaluation Tools*.
- Zhu, Y., et al. (2024). *Temporal Generalization in Large Language Models: The FreshBench Framework*.

Survey Focus

This state-of-the-art survey identifies the need to integrate **temporal dynamics** into the study of **political bias** in LLMs.

Thank You!

Questions?

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